

The Impact of ESG Technologies on Sustainable Business Performance

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ABSTRACT

In recent years, digitalization has been developing and there has been a strong drive towards environmental, social and governance (ESG) responsibility as well. These both things happened at the same time, for ESG accountability, businesses are using digital technologies. So the question is: How do these digital technologies translate into sustainable business performance? This study is specific to the EV industry, and investigates just this one question: how exactly can AI data systems and digital tracking tools make companies more sustainable and better-performing? Electric vehicle industry selected to have high expectations of environmental performance, tight government regulations and fast technological development. We conducted a search on Google Scholar in 2021–2026 using three Boolean strings following the guidelines of PRISMA 2020.

A total of 25 studies passed the title/abstract screening and the full-text screening phase from the 113 records identified. All have been quality-appraised and analysed using a three-stage thematic synthesis and results were interpreted using four theoretical lenses (stakeholder, institutional, natural-resource based and legitimacy theory). Our research found out that ESG technologies like AI and other digital tools help the companies to monitor their environmental impact such as pollution they have caused energy they have used and the battery life. The companies can use blockchain and battery passports to trace the material and its source, making the supply chain transparent and responsible.

AI helps to create better ESG reports, but it also helps companies avoid making false claims, or over-hyping their environmental measures, which is known as “greenwashing.” But all that can only be achieved if the company has the right skills, systems and strategy in place to implement ESG technologies and thereby improve innovation, business stability and also the financial results. One thing that this study does have is that it is geographically biased, as China is the leader in the EV industry worldwide, so the results may not necessarily be applied to the listed companies in all countries.

The research is particularly focussed on the electric vehicle industry. Describes the importance of ESG technologies, or technologies that are used by companies to improve their sustainable performance. It also sets out an agenda for future research.

Keywords: *ESG technologies; sustainable business performance; electric vehicles; systematic literature review; environment; social; digital transformation; corporate governance*

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1. Introduction

Over the last ten years the world has seen two major shifts in business that have altered the way that the businesses have been run. The first is the tech boom that ushered in digital transformation and at the same time, the second is sustainability itself that propelled towards ESG accountability. Since there was a rise in sustainability, it meant that the companies are no longer judged just on the basis of their profitability, now the investors, governments and even the public wanted proof that the business is protecting the environment, treating the people fairly and running the business ethically.

Now digital transformation was also required since the businesses operate in a highly dynamic environment and new powerful tools like AI, the Internet of Things (IoT), blockchain, big-data analytics were being used to measure and manage almost everything they do, businesses adopted them to measure their ESG performance as well. Typically, these are considered entirely separate issue areas: sustainability and the tech boom. Where they intersect — and when and how — the digital tools a company chooses to track and manage its ESG impacts are an asset to sustainable business performance is less clear.

There's already a substantial body of work that shows that companies with strong environmental and social practices are more financially successful – but only when sustainability is integrated, not added as a superficial PR stunt or a last minute edition, or just put on the shelves and forgotten, but deeply ingrained in the company's culture and operations. However, the link between ESG results and firm value provides very little information about the channels a firm uses to achieve its results. It is in the instrumental space such as the preventive dashboards which helps the company in real time in cases like when a machine starts emitting too much pollution, the dashboard identifies it even before it harms the environment and results into a massive government fine, the distributed Ledger using blockchain which tracks the exact journey of a raw material from the extraction to the factory making the supply chain completely auditable and proving the materials are ethical, the AI truth detector helps to read and analyze companies' public statements and the financial disclosures etc. over any number of years and identify the inconsistency and prove whether they're actually keeping their climate promises or not.

Electric vehicles industry is a key area in this study, as it is the ideal real world example of ESG (Environmental, Social and Governance) technology. A few main reasons why EV industry was chosen are that EVs face new environmental challenges as EVs themselves do not pollute the environment when driven but building them creates other environmental problems, risky supply chains as EV parts come from all over the world, it is very hard to track where materials come from or there are known human rights and unfair labor issues tied to mining those materials, EVs also have the tech advantage since they are highly computerized and automakers already have the digital tools like data tracking and sensors required to monitor and report on these environmental and social issues.

The EV industry is one of the hardest and most critical environmental and social industries to navigate, yet it also has access to the data technology necessary to overcome those challenges, but it also has the potential to be a fascinating test ground – where ESG technologies and data should be – if they are at all. Another reason for our motives is related to methodology. Earlier research on this matter has often confused technologies with their results. When a corporation is reported to be using digital ESG tools, it is assumed to be achieving sustainable objectives, which is not always possible.

In this study, clear steps of research have been taken. We make a clear distinction between the company making use of certain technology and the results it has achieved.

Objectives of the Study

This study aims to investigate the influence of Environmental, Social and Governance (ESG) technologies on business operations to achieve sustainability with the focus on the electric vehicles sector. The specific objectives of the study are:

- To investigate how digital ESG technologies (such as Artificial Intelligence, Industrial IoT and Big Data Analytics) directly impact corporate ESG performance and sustainable business outcomes in the EV industry.
- To study the mediating factors that include green product innovation, information transparency and corporate risk taking affecting the translating of digital transformation into tangible sustainability results.
- To establish the strategic facilitators and barriers (that comprise features of the top management team, organizational digital culture, etc.) for ensuring the successful adoption of ESG technologies.

The rest of the paper is structured as follows:

The literature review of ESG technologies and sustainable business practices is conducted in section 2. The research framework, data collection methods, etc. are outlined in section 3. Data analysis is presented in section 4, along with discussion of the results and conclusions. The research findings are summarised in section 5 along with the prospects for future studies and recommendations.

2. Literature Review

A common complaint about writing about ESG-technology is that it is observing but not explaining. In order to prevent such, the synthesis is read as per four complementary theories. These address different questions on the benefits of adopting a digital ESG tool for an EV company and whether and how it can help—or hinder—company performance. They both provide the causal vocabulary that is missing from a purely descriptive account.

2.1 Theoretical and Conceptual Framework

But to see how digital tools really impact business performance, it's important to examine the reasons for using them. Four main theories are used to explain this process in this study. The first one is the stakeholder theory (Freeman, 1984). It is not enough for a business to be just a money spinner for shareholders, a business must provide value for the government, its customers and its local communities. These groups are extremely vocal for EV companies. Individuals are asking themselves about the ethics of the batteries and regulators want to monitor emissions.

Digital ESG tools enable the company to gather data that is easy to collect and demonstrate to these groups that the company is being responsible. Institutional theory (DiMaggio & Powell, 1983) is the second concept. This theory presents that companies will be the ones to follow new practices not only for efficiency, but also because of external pressure. Governments are formulating stringent environmental regulations and there's a significant impetus to follow successful competitors in the EV segment. This also involves firms embracing AI and IoT solutions to adhere to regulations and expectations. But the theory has also got its haters, as others have feared that companies would

purchase the technology with a view to window dressing rather than changing their business practices.

In order to ensure that the technology will actually help performance, the resource-based and natural-resource-based perspectives (Barney, 1991; Hart, 1995) are explored. These theories propose a company's competitive advantage is derived from having unique resources that are difficult for others to replicate. These are special resources such as digital capabilities like big data analytics or blockchain tracking system. Utilizing these digital tools in a targeted way to reduce pollution and improve resource management creates a strong in-house capability which results in long-term financial success for the automaker. The legitimacy theory (Suchman, 1995) is the last theory. In other words, a company has to demonstrate that its acts are acceptable to the society. The public is quite skeptical of any climate promises at this time. Thus the trustworthiness of ESG data is of utmost significance.

AI truth detectors and secure data platforms are examples of technologies that help prevent greenwashing and ensure that when the public says it, it means it. This will preserve the company reputation and its business. The four theories are all combined to create the conceptual framework for this study. Digital ESG technologies are being forced on the EV companies by outside pressures from government and public. These tools, when adopted, offer real time monitoring and supply chain transparency.

These tools will enhance environmental monitoring, social equity and corporate governance, provided that the company implements them in the appropriate manner. In the end it translates to sustainable business performance, the company becomes more profitable, while they protect the planet at the same time. 21

2.2 The Role of Digital Technologies in ESG performance

Digital technologies are transforming businesses' approaches to their environmental footprint. Factories have real-time visibility of their operations thanks to the IoT and AI integration (Román-Salinas et al., 2026). For instance, AI and IoT technologies can collaborate to anticipate machine failure before it occurs and optimize energy consumption much more intelligently (Mansour et al., 2025a). The real time data feedback can help energy intensive manufacturing facilities to pinpoint the source of their energy consumption and enable a more efficient utilization of resources within the framework of the circular economy (Ma et al., 2024).

By applying these tools in their practice, EV manufacturers can greatly diminish their overall carbon footprint, and transform their sustainability objectives into tangible green innovation outputs (Wang et al., 2026). These technologies also have a sizable impact on the social performance of a business. Social sustainability is the aspect of a company that treats its employees fairly and ensures that everyone the company engages with—the production chain—in is being treated in a responsible way.

With the support of big data analytics, a company can analyze huge amounts of supplier data, ensuring that they comply with ethical and green standards (Cheng et al. 2025). The synergy of green strategies and big data can enhance the overall sustainability of business operations and create a safer workplace (Aslam, 2026). Moreover, the shift towards a more human-centric approach in manufacturing is enabling the use of IoT devices to assist workers in providing more ergonomic

monitoring and safer working systems (Román-Salinas et al., 2026). This forms an open network that safeguards employees and local communities. Last but not least, digital technologies are fundamentally transforming the landscape of corporate governance and reporting on corporate achievements. Honesty and good communication with the public are essential to good governance.

The company information becomes extremely transparent due to the big data infrastructure, which in turn introduces more external monitoring, and compels the company to take action in line with the ESG initiative to safeguard its reputation (Wang & Zhang, 2024). AI then goes one step further by automating data gathering and eliminating human error from sustainability reports (Alshareef, 2025). Recent research has shown that generative AI can dramatically improve the quality of information revealed by a company making it hard to argue against the existence of greenwashing (Xu et al., 2026). Transparency with regards to the data on AI and ESG leads to trust with investors and promotes sustainable financial value (Bogdan et al., 2025).

2.3 Challenges in Technology Adoption

The implementation of digital tools is not simple, although it has great advantages. One of the major challenges that many companies face is the financial expenses. It is essential to invest hugely in infrastructure and constantly train staff to create a robust digital infrastructure. The research demonstrates that high adoption costs, lack of digital-skills and infrastructure issues prevent many manufacturers from leveraging big data analytics effectively (Cheng et al., 2025).

This is a significant issue for small and medium sized businesses. Adopting advanced solutions such as generative AI is very costly and tends to be a game for big companies as it leaves smaller suppliers behind (Xu et al., 2026). Moreover, the limited supply of resources is a recurring issue faced by companies in developing areas, making it extremely difficult to manage high costs and low profitability in the implementation of digitalization (Aslam, 2026). In addition to the cost, there are significant technical challenges. The complexity of the factory's operation makes it difficult to integrate the use of old physical equipment with new AI and IoT sensors, which can create dynamic uncertainties that prevent real-time data feedback from being retrieved (Ma et al., 2024).

When combining different data types from multiple sources, there are generally interoperability problems because the various systems are unable to communicate properly with one another (Román-Salinas et al., 2026). Moreover, AI and IoT integration not only introduces new risks to the business but also opens the door to serious data security risks (Mansour et al., 2025a). With a huge amount of sensitive data collected regarding the supply chain and environment, protecting this data from cyber attacks is a serious concern.

Last but not least, there is the possibility of "show" technology. In the absence of a robust ethical strategy, digital tools can even camouflage unethical practices. Often times, the decisions made by complex AI algorithms are opaque, that is, no one can see the model's decision-making process or how the data is being processed (Xu et al., 2026). The opacity is conducive to symbolic environmental action by companies, rather than to real, substantive change. In order to be effective, technology needs to be embraced by the top leadership and have a culture of accountability and data integrity.

2.4 Research Gap

Digital transformation and sustainability have been separately studied in the past and, when studied together, sustainability and digital transformation are often intertwined with the adoption of digital tools rather than with environmental and/or social impacts. One key issue in the literature has been that companies being 'made' sustainable by merely introducing an AI or IoT system was not actually confirmed whether these systems actually lower emissions or enhance working conditions. Moreover, existing research is largely based on general manufacturing information and does not take into account the highly regulated nature of the EV industry and specific issues such as battery lifecycle management, ethical sourcing of minerals, etc. To fill these gaps the study is limited to the EV sector, while ensuring that the technologies acquired by companies are separated from the sustainability impacts they generate. The results of this research demonstrate that ESG technologies support sustainable business performance only when they include the appropriate organizational strategy and effective internal governance, as they are able to influence the exact mechanisms that enable their success, such as green product innovation and transparency in the supply chain.

3. Methodology

The review was carried out and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 statement (Page et al., 2021). Due to the study designs and the absence of a single effect size reported across the different studies in the evidence base, a quantitative meta-analysis was not attempted; it is qualitative and thematic in nature. The items below refer to the PRISMA 2020 which are followed by the steps below, and a completed checklist will be available from the authors.

3.1 Protocol and registration

Prior to screening a brief internal protocol was prepared, which included questions, eligibility criteria, search strings and the synthesis approach. This is noted as a limitation in the review process because the review was not registered on PROSPERO which usually does not accept management and business reviews. No amendments were made to the eligibility criteria after screening started.

3.2 Eligibility criteria

Studies were included if they were peer-reviewed, published between January 2021 and September 2026, written in English and used at least one digital ESG technology to address a sustainability, ESG or business-performance outcome, relevant to the EV or broader automotive-manufacturing sector, either directly or indirectly, with evidence that was transferrable to firm-level in the automotive-manufacturing sector. Those engineering studies where there was no sustainability/performance linkage, those on unrelated sectors with no transferable insight, non-peer-reviewed grey literature as primary evidence, and opinion pieces which lack substantive analysis were excluded. The criteria are given in full in Table 1.

Table 1. Inclusion and exclusion criteria

Dimension	Inclusion	Exclusion
Time frame	Published Jan 2021 – Sep 2026	Published before 2021
Source type	Peer-reviewed journal articles and rigorously reviewed conference papers	Blogs, editorials, theses, non-reviewed reports (used only as background context)
Language	English	Non-English with no English full text
Topical focus	At least one digital ESG technology (AI/ML, IoT, blockchain, big data, digital twins, ESG platforms) linked to an ESG/sustainability/performance outcome	Technology studies with no ESG or performance linkage; ESG studies with no technology component
Sector relevance	EV/automotive sector, or firm-level evidence transferable to the EV context	Sectors with no transferable relevance to EV firms
Contribution	Reports empirical results or a structured review with analysis	Descriptive notes/opinion with no analysis

AI/ML = artificial intelligence / machine learning; IoT = Internet of Things.

3.3 Information sources

In accordance with the design of the review, Google Scholar was the main source of information used. It has a wide spectrum of journals, disciplines and early-access articles, which are appropriate for a new, cross-disciplinary research area, and it brings to light finance, management and engineering forums where such literature can be found. The last search date of the database was 6 September 2026. The database search was complemented by backward citation search of included studies and targeted consultation of organisationally available sources (International Energy Agency and similar organisations) for contextual information to minimise the risk of missing important work.

3.4 Search strategy

Three concept blocks (ESG/sustainability, digital technology and the EV/automotive context) were merged with performance terms in three Boolean strings. Total number of records identified was kept below 200 to ensure screening remained rigorous, reproducible, and strings were iteratively refined for a balance of precision and recall. The following strings were passed to the functions in table 2, and the strings returned were listed.

Table 2. Search strings (Google Scholar) and records identified

#	Search string (title, abstract and keywords)	Records
S1	allintitle: ("ESG" OR "sustainability") AND ("IoT" OR "Internet of Things" OR "big data") AND ("manufacturing" OR "electric vehicle")	39
S2	allintitle: ("digital transformation") AND ("ESG performance" OR "green innovation") AND ("firm performance" OR "new energy vehicle")	14
S3	allintitle: ("ESG" OR "disclosure") AND ("artificial intelligence") AND ("greenwashing" OR "governance" OR "reporting quality")	60
	Total records identified (before de-duplication)	113

Search last run 6 September 2026. Wildcards () capture morphological variants. Counts are the records screened from each string; overlap across strings was removed at de-duplication.*

3.5 Selection process

Records have been exported and deduplicated resulting in 18 duplicates being removed due to overlap from the three strings, leaving 95 unique records. Titles and abstracts were then reviewed to make sure they met the eligibility criteria, and the full texts of potentially relevant reports were evaluated. The review team operating under the common protocol screened using the common screening criteria and resolved disagreements through discussion and referral back to screening criteria, with a conservative approach – keeping records at the title-abstract level where eligibility was not clear. The 95 unique records screened during these assessment phases were excluded, leaving 25 studies.

3.6 Data collection and data items

The method of structured extraction was used in taking essential information for every study in scope like bibliographic information; ESG technology or technologies studied; research techniques used as well as the context in terms of geography and area of study; ESG or performance outcome studied; nature of the relationship established; as well as the moderating, mediating, or limiting factors. A study which investigates some outcomes simultaneously, all of which also apply to review questions, was counted.

3.7 Quality appraisal

As the studies have a mix of qualitative, quantitate and review designs, a six-factor checklist was adapted from mixed methods evaluation to assess the quality of the studies (objectives clarity; appropriateness of design in relation to the problem studied; data adequacy and transparency; proper analysis; consistency of the evidence and conclusions made; importance of the results for the relationship between ESG and EV). Each study was ranked according to the quality criteria. While the appraisal influenced the interpretation of the studies recruited, it was not used to exclude any studies automatically because inclusivity is the priority in the systematic review in Section 4.3.

3.8 Synthesis method

The process of synthesizing findings has been done using Thomas and Harden's (2008) three-stage thematic synthesis. This involved three steps such as free line-by-line coding, organization of codes into descriptive themes, and abstraction into higher-order analytical themes that answer the questions of the review. This led to the emergence of four analytical themes in Section 4.4. This method was chosen instead of meta-analysis since the studies do not have common related measures nor similar designs and instead of narrative since it brings transparency and auditability to the analysis.

4. Data Analysis

4.1 Study selection

The searches led to the retrieval of 113 records through Google Scholar and relevant organizations search. After eliminating the duplicates (18 the same records were found in different searches), the 95 records have been screened for eligibility on the title and abstract level. In the course of screening, 70 records have been eliminated for various reasons—primarily, because they were not related to electric vehicles or automakers or did not relate digital ESG technology to sustainability. Thus, the rigorous selection led to the determination of 25 database studies which meet all eligibility criteria.

4.2 Characteristics of included studies

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Table 3. Characteristics and quality appraisal of the 25 included studies

Study	Technology	Design & Framework	Key findings	IoT/GPS integration enables real-time data feedback to successfully	Qual.
Ma et al. (2024)	Edge-cloud, IoT, & CPS	Empirical case study; Ceramic manufacturing survey; 212 employees	Digital transformation and green practices improve performance positively and mediate the impact of IoT-driven predictive maintenance, quality precision, and energy management.	optimize energy consumption and resource recycling	M
Khakwani et al. (2024)	Digital transformation	Conceptual review & multi-case; (PLS-SEM)	Time-based performance and IoT-driven predictive maintenance, quality	Time-based performance and IoT-driven predictive maintenance, quality	M
Mansour et al. (2025a)	AI & IoT (AIoT)	Manufacturing A-share listed companies	AI adoption and digital transformation mediate how	precision, and energy management.	
He (2024)	AI & Digital transformation	Survey & (2010–2021); 162 Italian manufacturing firms	ESG performance enhances IoT solutions facilitate energy efficiency and sustainability goals, contributing to firm participation	ESG performance enhances IoT solutions facilitate energy efficiency and sustainability goals, contributing to firm participation	H
Scarpino et al. (2026)	Industrial IoT (IIoT)	Panel: 21,064 obs. Chinese A-share (2010–2021)	Digital transformation positively moderates the ESG-firm performance link via efficient innovation.	contributing to firm participation	M
Zheng & Bu (2024)	Digital transformation	Panel: 21,064 obs. Chinese A-share (2010–2021)	Big data capabilities significantly enhance ESG performance, mediated by risk-taking and moderated by top management education.	Big data capabilities significantly enhance ESG performance, mediated by risk-taking and moderated by top management education.	H
Aslam (2026)	Big data analytics	Panel: 30,448 obs. Chinese A-share (2010–2021)	Digital transformation enhances ESG performance, mediated by risk-taking and moderated by top management education.	Digital transformation enhances ESG performance, mediated by risk-taking and moderated by top management education.	H
Sang et al. (2024)	Digital transformation	Panel: 30,448 obs. Chinese A-share (2010–2021)	Big data capabilities positively impact top-line sustainability, mediated by green supplier integration.	Big data capabilities positively impact top-line sustainability, mediated by green supplier integration.	H
Cheng et al. (2025)	Big data analytics	Panel: 27,254 obs. Chinese A-share (2009–2023)	Digital transformation strengthens ESG performance by reducing information asymmetry and improving internal governance, intensifying external monitoring.	Big data analytics helps mediate the impact of ESG on green innovation.	H
Wang et al. (2026)	Digital transformation	Panel: 27,254 obs. Chinese A-share (2009–2023)	Big data infrastructure enhances ESG performance by increasing transparency and governance, intensifying external monitoring.	Big data infrastructure enhances ESG performance by increasing transparency and governance, intensifying external monitoring.	H
Zhou et al. (2025)	AI	Panel: 1,418 Chinese manufacturing firms (DiD)	Big data infrastructure enhances ESG performance by increasing transparency and governance, intensifying external monitoring.	Big data infrastructure enhances ESG performance by increasing transparency and governance, intensifying external monitoring.	H
Wang & Zhang (2024)	Big data infrastructure	Panel: 1,418 Chinese manufacturing firms (DiD)	Big data infrastructure enhances ESG performance by increasing transparency and governance, intensifying external monitoring.	Big data infrastructure enhances ESG performance by increasing transparency and governance, intensifying external monitoring.	H
Sari et al. (2025)	AI	Machine learning empirical model	IoT technologies effectively support circular economy and socio-environmental practices	IoT technologies effectively support circular economy and socio-environmental practices	H
Cavalieri et al. (2022)	IoT	Conceptual review	IoT technologies effectively support circular economy and socio-environmental practices	IoT technologies effectively support circular economy and socio-environmental practices	M
Salma et al. (2025)	AI & IoT auditing	Empirical (PLS-SEM); 60 firms, 120 cases (Automotive)	AI and IoT improve ESG compliance and tracks Scope 3 emissions, though introducing energy trade-offs.	AI and IoT improve ESG compliance and tracks Scope 3 emissions, though introducing energy trade-offs.	H
Hu et al. (2022)	Green Industrial IoT	Structural equation modeling; Manufacturing	Identifies the external regulatory and internal efficiency drivers that govern Green IIoT adoption for eco-sustainability.	Identifies the external regulatory and internal efficiency drivers that govern Green IIoT adoption for eco-sustainability.	M

Zhang & Lei (2025)	IoT & Big data	Optimization algorithms; Industrial supply chains	Algorithmic resource balancing guided by IoT sensors delivers quantifiable ESG compliance gains.	H
AL-Shboul (2024)	Big data analytics	Covariance-based SEM; Supply chain manufacturing	Big data capabilities build persistent competitive advantages when channeled into sustainable sourcing.	H
Román-Salinas et al. (2026)	Industrial IoT (IIoT)	PRISMA 2020 Systematic Review; Production systems	Establishes mechanistic links between IIoT architecture and manufacturing energy/resource efficiency.	H
Wang & Zhang (2025)	Digital transformation	Structural equation modeling; Industrial enterprises	Internal firm integration and organizational digital culture moderate how digital transformation determines ESG performance.	H
Bogdan et al. (2025)	AI governance & disclosure	Empirical disclosure index; Industrial reporting	Automated ESG disclosures and AI reporting quality significantly improve external ESG scores.	H
Alshareef (2025)	AI	Panel analysis; Corporate disclosure systems	AI integration in sustainability reporting curtails selective disclosure, improving the reliability of ESG reports.	H
Huang & Xu (2026)	AI	Empirical panel; Heavy industrial enterprises	AI implementation improves mandatory environmental transparency by enhancing internal audit speeds.	H
Mansour et al. (2025b)	AI	Dynamic panel analysis; Industrial emerging markets	AI adoption amplifies green innovation efficiency and optimizes ESG governance, minimizing emissions abatement costs.	H
Sætra (2023)	Responsible AI	Theoretical and governance modeling	Establishes a foundational protocol for evaluating and auditing the socio-environmental implications of AI-driven ESG systems.	M

Qual. = quality appraisal rating (*H* = High, *M* = Medium); *PLS-SEM* = Partial Least Squares Structural Equation Modeling; *DiD* = Difference-in-Differences; *obs.* = observations.

4.3 Quality of the evidence

Not surprising, given that the corpus is from peer-reviewed venues and filtered for analytical content. Weaknesses were also fairly consistent, as they were poorly worded questions, unclear objectives, or endogeneity issues that were not properly addressed in the quantitative studies (such as with instrumental variables, difference in differences and staggered adoption designs). The narrower weaknesses are the following: Several studies conclude by studying a broad sample of listed firms, not EV firms; the measurement of what is called “digital transformation” varies from patent counts to text-mined indices, making it difficult to compare the studies; and the qualitative or design-science studies provide valuable insights but rely on small or single cases. These patterns are used to moderate the confidence of the individual claims and are reiterated in the limitations.

4.4 Thematic synthesis

Four analytical themes were generated from coding. The first two are about the environment and social dimensions and the technology that supports them; the third is about the governance and integrity of ESG information; and the fourth is about the strategic conditions that will create a success rate for each. The links between technology and applications and the resulting outcomes are summarised in Table 5, while the themes and their related theory are mapped in Table 6.

4.4.1 Theme 1 — Environmental intelligence

The evidence is most vast in what can be described as environmental intelligence: the application of AI, Industrial IoT and edge-cloud collaboration for viewing and guiding a company's environmental impact in real-time. In the heavy manufacturing and EV application, the concentration is on energy management and equipment life cycles. AI and IoT (AIoT) combine to make significant advances in predictive maintenance and smart energy management (Mansour et al., 2025a). Edge-cloud cooperation combined with Cyber-Physical Systems (CPS) enables data feedback in real time, which enables optimization of the energy use of equipment and resource recycling, following the principles of the circular economy (Ma et al., 2024; Román-Salinas et al., 2026). The literature, however, is cautious when discussing the pathway: sustainable performance is increased with IIoT and big data capabilities, but in turn, the results depend heavily on the level of participation in the firm and the internal efficiency drivers (Scarpino et al., 2026; Hu et al., 2022). The theme is perfectly suited to the natural-resource-based thinking view: The resource that can lower pollution is sensing and analytics, but the value comes from the work routines that use the data to drive action.

4.4.2 Theme 2 — Traceable and responsible value chains

The second theme is a collection of technologies that seek to transparency the supply chain, while the EV industry's strongest ESG exposure is in the upstream. When leveraging big data analytics for the purpose of sustainable sourcing and green supplier integration, the result is persistent competitive advantages are demonstrated (AL-Shboul, 2024; Cheng et al., 2025). In terms of social dimension, research on the impacts of digital supply chain transformation on ESG performance has clearly emphasized the importance of internal firm integration and an organizational digital culture (Wang & Zhang, 2025). Additionally, algorithmic resource balancing, based on IoT sensors, ensures

measurable ESG gains in industrial supply chains (Zhang & Lei, 2025). These are technologies that transform opaque and adversarial supplier relationships into auditable, accountable supplier relationships, as described in stakeholder theory.

4.4.3 Theme 3 — Governance, disclosure integrity and anti-greenwashing

The third theme is also the one where the most interesting recent research comes to light, and where the tool–outcome distinction is most salient. A series of studies reveal that AI implementation enhances corporate information governance which leads to better ESG performance, specifically by increasing information transparency (information asymmetry) and the quality of information disclosure (Zhou et al., 2025; Bogdan et al., 2025). Most obviously, AI-powered detection systems are found to reveal the symbolic versus substantive nature of ESG reporting and actively prevent ESG greenwashing (Sari et al., 2025). AI integration in sustainability reporting also helps to greatly reduce selective disclosure and increases forced environmental transparency by boosting the speed of internal audit (Alshareef, 2025; Huang & Xu, 2026). Legitimacy theory interprets these instruments as a public infrastructure for bridging the say–do gap, to make public environmental claims more accountable to data.

4.4.4 Theme 4 — Strategic integration, performance and barriers

The final theme weaves through the others, and is a key qualification of the review: the payoff from ESG technologies is contingent, not guaranteed. Digital transformation is a reliable driver that reinforces the ESG–performance connection, but it is also an enabler, as it enhances a correlation that is heavily dependent on green innovation to make a difference (Zheng & Bu, 2024; He, 2024; Wang et al., 2026). The institutional dimension comes to the fore when it is demonstrated that the digitalization capability of a firm plays a crucial role in translating a commitment to sustainability into real sustainable performance, where the initial mediator is operational sustainability (Aslam, 2026). Moreover, human and strategic conditions are significant boundary conditions, with corporate risk-taking being a major determinant of the success of digital transformation in boosting ESG performance and the educational level of the top management team being a moderating condition. This is the same across the corpus – the use of technology for compliance.

The final theme goes beyond these senses and is part of the core qualification of the review: the payoff from ESG technologies is not guaranteed, but contingent. Digital transformation can consistently enhance the ESG–performance relationship, but it is merely an enabler, as it is supported by the strong reliance on green innovation for its success (Zheng & Bu, 2024; He, 2024; Wang et al., 2026). The institutional dimension comes to the fore through the finding that the institutional ability of a firm affects its performance in delivering on its sustainability commitments, with operational sustainability as the key intervening variable (Aslam, 2026). In addition, human and strategic conditions are significant boundary conditions, as the ESG performance improvement through digital transformation is significantly mediated by corporate risk-taking, and moderated by the educational level of the top management team (Sang et al., 2024). The application is the same across the corpus: Technology is just used to meet requirements.

Table 5. Key ESG technologies, EV-sector applications and reported outcomes

Technology	Representative EV-sector application	Reported outcome	Primary pillar
AI / machine learning	Emissions analytics; greenwashing detection; internal auditing	Higher ESG scores; better disclosure quality; reduced greenwashing	E / G
IoT & edge-cloud systems	Real-time energy, emissions, and predictive maintenance monitoring	Efficiency gains via circular practices and reduced equipment downtime	E
Big-data analytics	Sustainable sourcing; supplier integration tracking	Improved supply chain transparency and competitive advantage	S / G
AIoT (AI + IoT Fusion)	Scope 3 emissions tracking; automated resource balancing	Quantifiable ESG compliance and advanced asset performance	E
ESG / digital platforms	Integrated reporting and automated transparency governance	Lower information asymmetry; auditable governance	G

E = environmental, S = social, G = governance. Outcomes summarise the direction of evidence in the included studies, not effect sizes.

Table 6. Mapping of analytical themes to theoretical lenses

Theme	What the evidence shows	Principal lens
1. Environmental intelligence	Sensing-and-analytics capability enables pollution prevention and stewardship, contingent on routines	Natural-resource-based view
2. Traceable, responsible value chains	Traceability tools convert opaque supplier ties into accountable stakeholder relationships	Stakeholder theory
3. Governance & disclosure integrity	AI/ledger tools raise disclosure credibility and curb greenwashing, managing the say-do gap	Legitimacy theory
4. Strategic integration & barriers	Adoption is driven by legitimacy pressure; performance depends on capability and integration	Institutional theory + RBV

5. Discussion

5.1 Interpreting the evidence against the review questions

To address the first research question, digital ESG technologies have a clear impact on sustainable business performance within the EVP, but not for all mechanisms. Assessing and managing assets precisely for environmental gains. Governance improvements occur as a result of improved and more trustworthy information. Financial gains are witnessed when governance is strong and technology is fully incorporated in the business strategy. The 25 studies in this corpus are insufficient to back the general argument that firms in general that implement ESG technologies simply perform better. Rather, they demonstrate that specific technologies have specific impacts on specific results, and they show how.

When it comes to the second question, it's about strategy. Digital transformation is the strongest cross-cutting finding: digital transformation is a moderator. It is a way of strengthening a link between an ESG and a performance that companies must first create. This is the applied and practical application of the resource based approach: the advantage is in the capability to embed the technology within the organization, not in the technology itself.

The enablers and barriers for the third question are very apparent. Strategic integration, organizational readiness, top management education, and supportive institutional environment. Challenges are high implementation costs, lack of common data standards, and the danger of 'showing up' to show they are adopting. The same institutional pressures that incentivize companies to use these tools can also lead to "lip service only" compliance if the company isn't really digitally capable.

5.2 Contribution and the gap addressed

This review has three major contributions. First, it brings together a rapidly emerging evidence base into a sector-specific synthesis for the EV industry. It supersedes anecdotal descriptions of big car companies with a focused presentation of 25 peer-reviewed studies. Second, it is strictly keeping the analytical distance between the tools of the ESG and the ESG outcomes, which is not maintained in a lot of the previous studies. The review achieves this, finding the right mechanisms, e.g. green innovation, and the boundary conditions that link the two together. Third, it provides solid theoretical underpinning by linking the evidence to theories of the stakeholders, institutions, natural resources, and legitimacy. The study does not only provide empirical evidence of the relevance of these technologies in the field of EV, it also provides the concepts of why and when.

5.3 Implications for practice and policy

Managers: apply the strategic concept to ESG technology and not merely a reporting cost. The economic benefits are for companies that use these tools to be more deeply embedded in their operations and governance. Instead, a company should use an emissions analytics platform in conjunction with the authority to influence real data-driven decisions, rather than just using it for appearances. Investment in the skills of employees and top management readiness is as crucial as the technology.

This review suggests that standardisation is the most promising intervention for policymakers. The issues of interoperability that emerge in various studies could be addressed by common rules and battery traceability standards. Moreover, implementing credible assurance requirements would

enable AI truth detectors to make a significant impact in preventing greenwashing. Early examples of such coercive yet enabling pressure are the EU Battery Regulation and digital battery passports.

5.4 Limitations

There are two restrictions in this study. As far as the evidence is concerned, the data is geographically biased towards the Chinese listed firms. China is the industry leader for EVs, making it difficult to generalize the results to other regulatory markets. Long-term effects are also not yet apparent in the field, as it is very young. Also, there are several studies that indirectly demonstrate their relevance to the EV space by utilizing a wide sample of manufacturing companies rather than just testing EV companies.

As to the review process, one main database was used. Google Scholar is appropriate for the study design, but has some disadvantages in terms of coverage compared to a multi-database search. The review focuses on English language work; it did not get registered on PROSPERO; and it is of a qualitative synthesis type, where interpretative judgment occurs during building the themes. These risks are minimised but not eliminated by the strict protocol and quality appraisal.

5.5 Future Research

There are four directions of action that stem from gaps identified. To begin with, there are no reliable firsthand findings for the field that confirm the relationships that have been hypothesized between EV research and development and manufacturing firms in general. First of all, there is no direct field evidence to confirm the hypothesized relationships between EV R&D and all manufacturing firms, but only evidence for those relationships based on general manufacturing firms. Second, longitudinal and panel designs should follow up to determine if early performance effects are either maintained or faded or strengthens as these technologies develop. Thirdly, the researchers should delve into the actual organizational capacities needed to foster the technology into performance, hopefully through qualitative case studies within EV companies. Fourth, with the rise of AI from measurement of ESG data to decision-making, the governance of AI becomes an important research area. There are multiple challenges that must be addressed, such as algorithmic bias, data security and the ethics of automated assurance, which have yet to be explored by the current sustainability literature, and which urgently require focus.

6. Conclusion

The aim of this systematic review was to identify digital ESG technologies and conditions for their use in supporting sustainable business performance in the challenging context of the electric vehicle industry. After reading through 4 theoretical lenses and reading 25 peer-reviewed studies since 2022, it concludes that these technologies are real enablers of sustainable performance, but selective and conditional. AI, IoT, big data analytics, and edge-cloud systems refine environmental monitoring, big data and IoT tracking systems make value chains traceable and responsible, and AI-driven governance tools increase the trustworthiness of environmental, social, and governance (ESG) information and increase the ability to prove greenwashing is reduced. However, a common theme emerges in every theme: value is created in the context of organizational capability and strategy that incorporates technology, and is reduced by cost, skill and standardization challenges. The message is evident in an industry that is increasingly being held to account for its sustainability claims, especially for an EV sector. What is no longer a question is whether to embrace ESG technologies, but whether a firm is able to develop the capacity to transform the technologies from disclosure tools into actual, verifiable performance tools.

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